Surya Partap Singh

Senior Data Engineer · Azure · Databricks · Spark · Lakehouse Platforms

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Summary

Senior Data Engineer with 5+ years building and scaling enterprise lakehouse platforms on **Databricks, Delta Lake, Apache Spark, and Azure** — unifying clinical, financial, and pharmacy data for analytics, reporting, and ML. Proven track record of optimizing distributed workloads, automating CI/CD pipelines, and mentoring engineering teams in high-stakes production environments.

Technical Skills

Programming: **Python**, PySpark, Spark SQL, SQL, C++
Data Platforms: **Databricks**, **Delta Lake**, Apache Spark, Hadoop, Hive, HDFS
Cloud & Storage: Microsoft Azure (Data Lake, Blob Storage, Azure SQL, ADF), Amazon S3
Streaming: Spark Structured Streaming, Delta Change Data Feed
Orchestration: Databricks Workflows, Azure Data Factory, Airflow
Data Quality & Governance: Great Expectations, schema evolution, anomaly detection, data lineage, observability
DevOps & IaC: GitHub Actions, Azure DevOps, ARM Templates, Git, Docker
Ways of Working: Agile, code reviews, mentoring, production support

Experience

Cencora — Dallas, TX
Senior Data Engineer, Data Platform — *Feb 2023 – Present*

- **Lakehouse Platform Architecture**: Designed and scaled a **Databricks** and **Delta Lake**-based medallion lakehouse that unified clinical, financial, and pharmacy data into reusable, governed datasets reducing time-to-insight for analytics, reporting, and data science teams by **50%**.
- **Spark Performance Optimization**: Improved end-to-end pipeline throughput by **30%** by tuning **Apache Spark** workloads across partitioning, shuffle behavior, join strategies, and cluster resource utilization.
- **Near-Real-Time Processing**: Reduced financial reporting latency from **24 hours to 15 minutes** by redesigning batch pipelines into incremental and near-real-time processing using **Delta CDF** and **Spark Structured Streaming**.
- **Unstructured Data Pipelines**: Built OCR-driven ingestion pipelines to extract and standardize data from unstructured patient documents, enabling curated datasets for clinical analytics and downstream ML workflows.
- **Data Quality & Observability**: Implemented automated data quality controls using **Great Expectations** and anomaly detection frameworks catching schema drift, freshness failures, and integrity issues before downstream impact, reducing data incidents by **35%**.
- **Platform Automation**: Established CI/CD and infrastructure automation for data pipelines and cloud resources using **GitHub Actions, Azure DevOps,** and **ARM Templates**, improving release consistency and cutting deployment time by **25%**.
- **Reusable Frameworks**: Developed reusable ingestion and transformation frameworks along with canonical data models, reducing onboarding time for new data sources by **40%**.
- **Cloud Cost Optimization**: Reduced Azure Databricks compute costs by **30%** through right-sizing cluster configurations, enforcing auto-termination policies, migrating low-priority workloads to spot instances, and consolidating redundant jobs delivering measurable cloud spend savings without impacting pipeline SLAs.
- **Unity Catalog & Data Governance**: Led adoption of **Databricks Unity Catalog** to enforce fine-grained access control, column-level security, and data lineage tracking across the lakehouse establishing a centralized governance layer that ensured regulatory compliance and audit-readiness across clinical and financial datasets.
- **ML & Data Science Enablement**: Engineered ML-ready Gold layer datasets and integrated **MLflow** experiment tracking into platform workflows — enabling data science teams to reliably reproduce experiments, accelerate feature development, and deploy models with well-defined, versioned data contracts.
- **Stakeholder & Cross-Functional Collaboration**: Partnered with data science, analytics engineering, and business stakeholder teams to define data contracts, consumption SLAs, and governance standards ensuring platform capabilities aligned with downstream reporting, ML, and self-serve analytics needs.
- **Technical Leadership**: Mentored a team of **3–5 engineers** on ETL design, Spark optimization, and platform best practices conducting structured code reviews, defining team-wide engineering standards, and reducing new engineer onboarding time by **30%** through reusable documentation and pair programming.

Cencora (Contract) Dallas, TX
Data Engineer *May 2021 – Feb 2023*

- **Production Data Pipelines**: Built and supported production ETL pipelines on **Databricks**, **Azure Data Factory**, and **PySpark** for large-scale clinical, pharmacy, and financial datasets.
- **Canonical Data Models**: Designed canonical, single-source-of-truth tables that standardized disparate source systems improving cross-team reporting consistency and eliminating **3+** redundant dataset copies maintained across business units.
- **Distributed Data Processing**: Optimized large-scale **Spark** transformations using broadcast joins, predicate pushdown, and partition-aware reads to improve runtime efficiency and reduce compute costs.
- **Workflow Reliability**: Automated and monitored ETL workflows through **Databricks Jobs**, implementing retries, alerting, and operational checks improving pipeline success rates to **99%+** and ensuring consistent SLA adherence.

Fuge Technologies Inc Dallas-Fort Worth, TX
Data Engineer *Feb 2021 – May 2021*

- **Batch Processing**: Built and optimized Hadoop and Hive-based batch pipelines for large structured datasets, applying partition pruning, bucketing strategies, and execution plan analysis to improve query performance in distributed environments.
- **Azure Data Ingestion**: Developed ingestion pipelines into Azure Data Lake using **Azure Data Factory**, supporting incremental loads, schema evolution handling, and downstream validation checks for analytics use cases.
- **Data Standardization**: Collaborated with cross-functional teams to normalize and standardize raw source data into consistent, analytics-ready formats, establishing naming conventions and structural patterns adopted across the pipeline.

Education

University of North Alabama Florence, AL
Bachelor of Science in Computer Science *Aug 2016 – Dec 2020*

- **Relevant Coursework**: Distributed Systems, Database Management Systems, Data Structures & Algorithms, Operating Systems, Cloud Computing, Software Engineering, Machine Learning